

E-VOTING

A PROJECT REPORT

For The Course

“Software Engineering: Mini Project – Stage I (CS3520)”

Submitted By:

Elangbam Flash (2301CS0303)

Marx Lourembam (2301CS0304)

Punshiba Oinam (2301CS0306)

3rd Year, 5th Semester

Department of Computer Science & Engineering

Guided By:

Dr. Hanjabam Saratchandra Sharma

MANIPUR TECHNICAL UNIVERSITY
TAKYELPAT, MANIPUR – 795004



A UNIVERSITY ESTABLISHED UNDER
THE MANIPUR TECHNICAL UNIVERSITY ACT, 2016



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MANIPUR TECHNICAL UNIVERSITY, IMPHAL

(A University established under the Manipur Technical University Act, 2016)

Imphal, Manipur- 795004

Website: - www.mtu.ac.in

DECLARATION

We, the students of Manipur Technical University, declare that the project titled **E-Voting** is an original work carried out by our team as part of our *Software Engineering: Mini Project - Stage I* course and has not been submitted elsewhere for any other degree or diploma. The content of this report is based on our understanding and the guidance provided by our supervisor. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

DATE: _____

PLACE: Imphal, Manipur

1. Elangbam Flash (2301CS0303)

2. Marx Lourembam (2301CS0304)

3. Punshiba Oinam (2301CS0306)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MANIPUR TECHNICAL UNIVERSITY, IMPHAL

(A University established under the Manipur Technical University Act, 2016)

Imphal, Manipur- 795004

Website: - www.mtu.ac.in

ACKNOWLEDGEMENT

We extend our heartfelt gratitude to Manipur Technical University for providing us with the platform to delve into the realm of software development.

Special thanks to our Head of Department for his invaluable guidance and support throughout this journey.

We would also like to thank our department faculty members for providing us with the foundational knowledge and skills necessary for undertaking this project.

Last but not least, we express our gratitude to each member of our team whose dedication and collaboration made this E-Voting project a reality.

DATE: _____

PLACE: Imphal, Manipur

Team Members:

Elangbam Flash

Marx Lourembam

Punshiba Oinam

Abstract

The project aims to:

- E-voting, or electronic voting, is a technological innovation designed to enhance the efficiency, accuracy, and accessibility of the electoral process.
- It leverages electronic systems, such as dedicated voting machines or online platforms, to record, store, and count votes securely.
- E-voting systems aim to mitigate traditional voting challenges, including human error, logistical complexities, and fraud.
- Key features include authentication mechanisms to ensure voter identity, encryption to protect vote privacy, and blockchain or other technologies for tamper-proof records.
- While e-voting promises transparency and faster results, it also raises concerns about cybersecurity, accessibility for all demographics, and trust in digital infrastructure.

TABLE OF CONTENTS

Section	Page No.
DECLARATION	i
ACKNOWLEDGEMENT	ii
ABSTRACT	iii
CHAPTER 1: INTRODUCTION	
1.1 Background	1
1.2 Objectives	1
1.3 Scope	1
CHAPTER 2: LITERATURE REVIEW	
2.1 Existing Systems	2
2.2 Challenges in E-Voting	2
2.3 Proposed Solution	2
CHAPTER 3: SYSTEM REQUIREMENTS	
3.1 Functional Requirements	3
3.2 Non-Functional Requirements	3
3.3 Technology Stack	4
CHAPTER 4: SYSTEM DESIGN	
4.1 Architecture	5
4.2 Database Schema	6
CHAPTER 5: IMPLEMENTATION	
5.1 User Registration	7
5.2 Login System	7
5.3 Poll Creation	7
5.4 Voting System	8
5.5 Results Display	8
CHAPTER 6: TESTING AND VALIDATION	
6.1 Functional Testing	9
6.2 Performance Testing	9
6.3 Security Testing	9
CHAPTER 7: CONCLUSION	
7.1 Summary	10
7.2 Future Scope	10
7.3 Limitations	10

Chapter 1: Introduction

1.1 Background

Voting is a fundamental right and an essential process in democratic systems. With the advancements in technology, traditional voting systems are gradually being replaced by electronic voting (E-Voting) systems to enhance efficiency, transparency, and accessibility.

1.2 Objectives

The primary objective of this project is to design and implement a secure, efficient, and user-friendly E-Voting platform that allows users to create polls, cast votes, and view results seamlessly.

1.3 Scope

This project covers:

- User authentication for secure access.
- Poll creation and management.
- Secure vote submission.
- Real-time results display.

Chapter 2: Literature Review

2.1 Existing Systems

Traditional voting systems rely on paper ballots, which are prone to manual errors and inefficiencies. Modern E-Voting systems aim to address these challenges but often face issues related to security and user trust.

2.2 Challenges in E-Voting

- Ensuring data security and integrity.
- Preventing unauthorized access and duplicate voting.
- Providing a seamless user experience.

2.3 Proposed Solution

This project leverages modern web technologies and a SQL Server database to create a secure and efficient E-Voting platform. By implementing strong encryption and authentication mechanisms, the system addresses the common challenges faced by E-Voting systems.

Chapter 3: System Requirements

3.1 Functional Requirements

- User Registration and Login.
- Poll Creation by authenticated users.
- Voting functionality with real-time validation.
- Results visualization.

3.2 Non-Functional Requirements

- **Scalability:** Support for a large number of users and polls.
- **Security:** Encryption of sensitive data.
- **Usability:** Intuitive user interface for all age groups.
- **Performance:** Quick response times for all actions.

3.3 Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js with Express.js
- **Database:** SQL Server 2014
- **Other Tools:** bcrypt for password hashing, CORS for cross-origin resource sharing.

Chapter 4: System Design

4.1 Architecture

The platform follows a **three-tier architecture**:

- **Presentation Layer:** Web interface for user interactions.
- **Application Layer:** Handles business logic and facilitates communication between the frontend and database.
- **Data Layer:** Stores user and poll data in SQL Server.

4.2 Database Schema

The database schema consists of three primary tables: **Users**, **Polls**, and **Votes**.

Table: Users

Column	Data Type	Constraints
id	INT	PRIMARY KEY, IDENTITY(1,1)
username	VARCHAR(100)	NOT NULL
email	VARCHAR(255)	NOT NULL, UNIQUE
password	VARCHAR(255)	NOT NULL

Table: Polls

Column	Data Type	Constraints
id	INT	PRIMARY KEY, IDENTITY(1,1)
title	VARCHAR(255)	NOT NULL
creator_id	INT	FOREIGN KEY REFERENCES Users(id)
created_at	DATETIME	DEFAULT GETDATE()

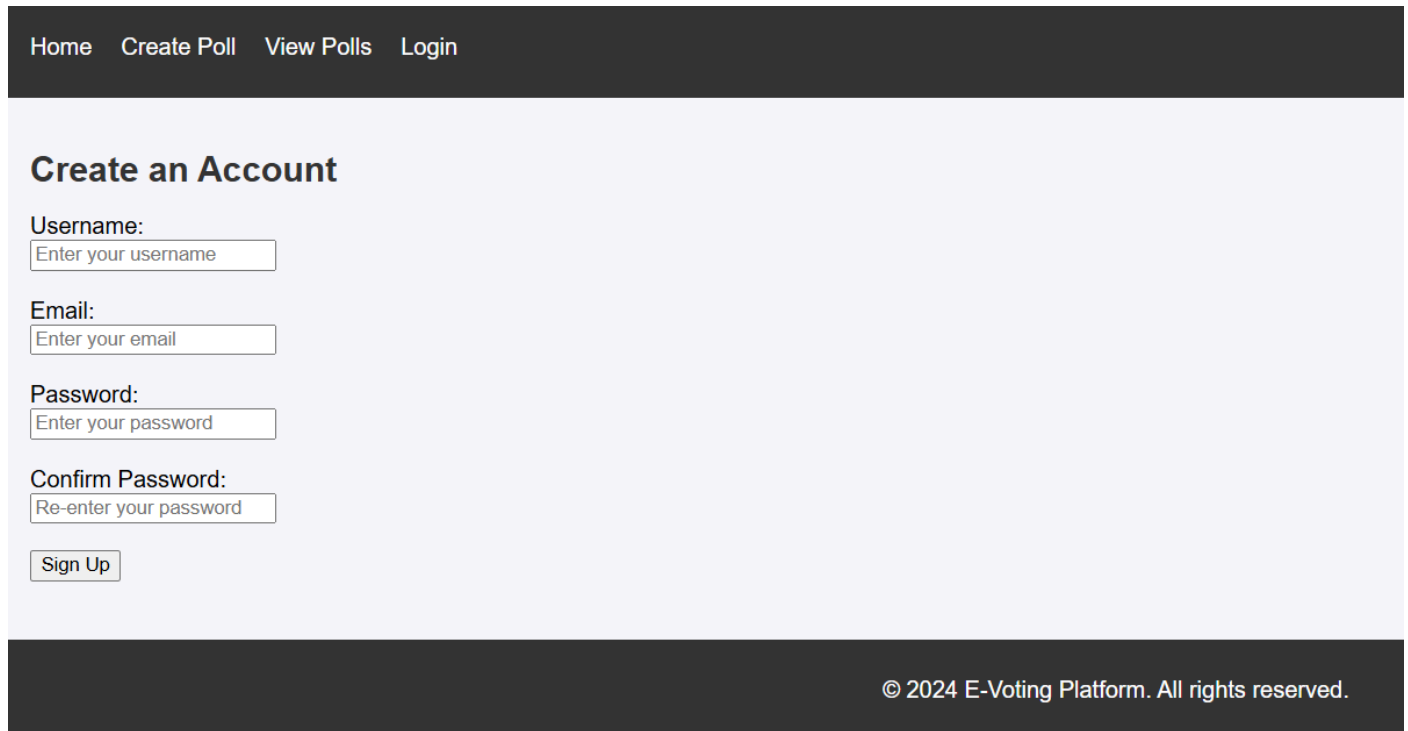
Table: Votes

Column	Data Type	Constraints
id	INT	PRIMARY KEY, IDENTITY(1,1)
poll_id	INT	FOREIGN KEY REFERENCES Polls(id)
user_id	INT	FOREIGN KEY REFERENCES Users(id)
choice	VARCHAR(255)	NOT NULL

Chapter 5: Implementation

5.1 User Registration

Users can sign up by providing their username, email, and password. Passwords are securely hashed using bcrypt before being stored in the database.

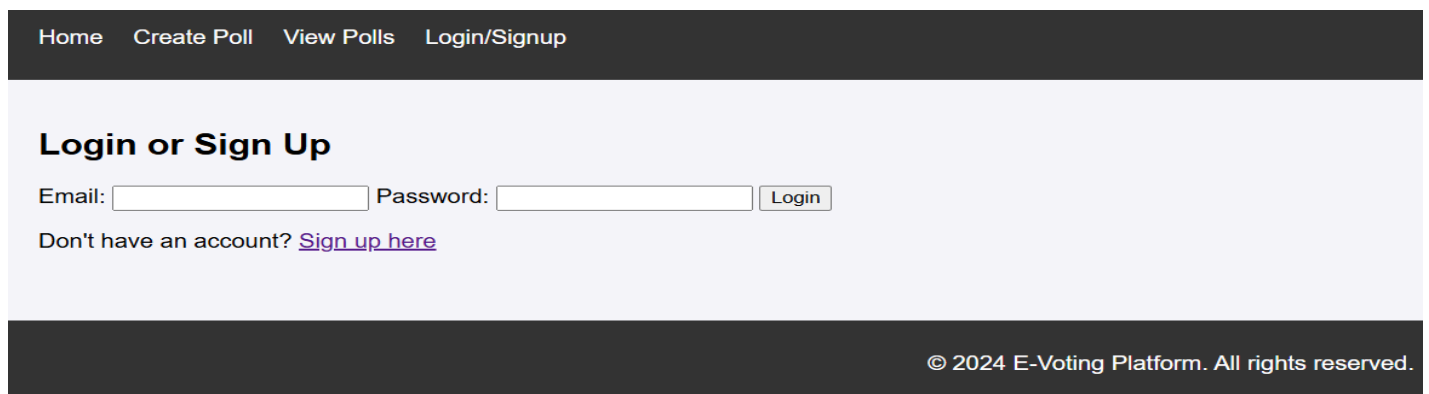


The screenshot shows a web application interface for user registration. At the top, a dark navigation bar contains links for 'Home', 'Create Poll', 'View Polls', and 'Login'. Below this, the main content area has a light blue background and is titled 'Create an Account'. It contains four input fields: 'Username:' with placeholder text 'Enter your username', 'Email:' with placeholder text 'Enter your email', 'Password:' with placeholder text 'Enter your password', and 'Confirm Password:' with placeholder text 'Re-enter your password'. A 'Sign Up' button is located below the input fields. At the bottom of the page, a dark footer bar displays the copyright notice '© 2024 E-Voting Platform. All rights reserved.'

Fig 5.1 Sign up page

5.2 Login System

Users can log in using their registered email and password. Authentication is performed by comparing the hashed password stored in the database with the entered password.

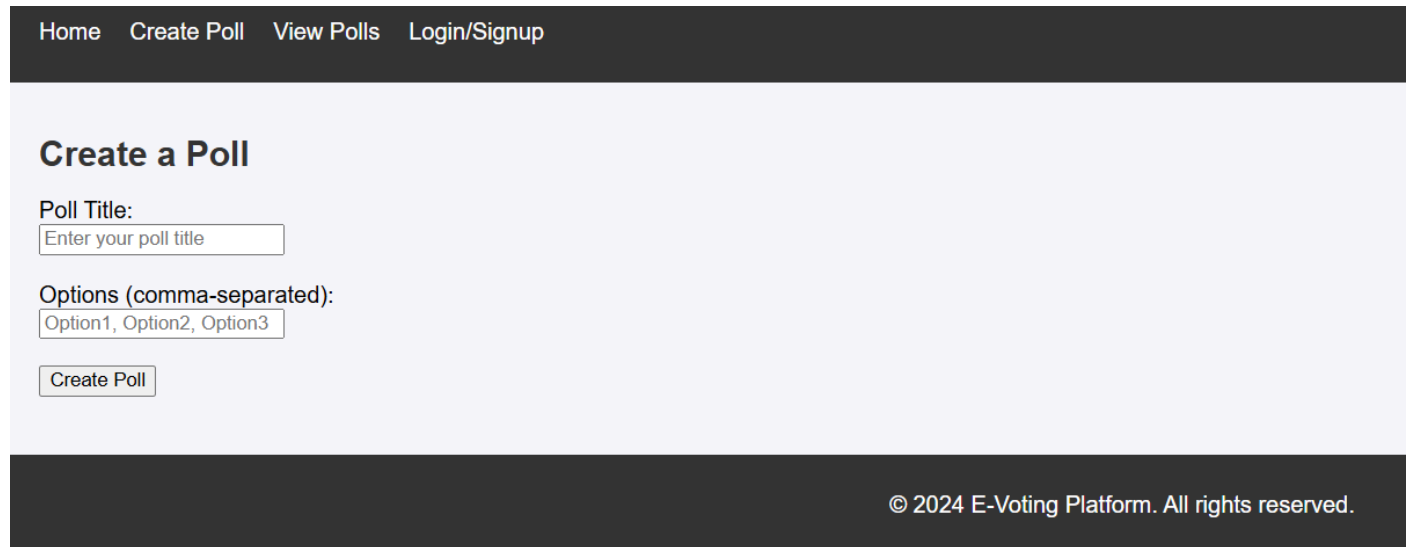


The screenshot shows a web application interface for user login. At the top, a dark navigation bar contains links for 'Home', 'Create Poll', 'View Polls', and 'Login/Signup'. Below this, the main content area has a light blue background and is titled 'Login or Sign Up'. It contains two input fields: 'Email:' and 'Password:', each followed by a 'Login' button. Below the input fields, there is a link that says 'Don't have an account? [Sign up here](#)'. At the bottom of the page, a dark footer bar displays the copyright notice '© 2024 E-Voting Platform. All rights reserved.'

Fig 5.2 Login page

5.3 Poll Creation

Authenticated users can create polls by specifying a title. Polls are associated with the creator's ID and timestamped.

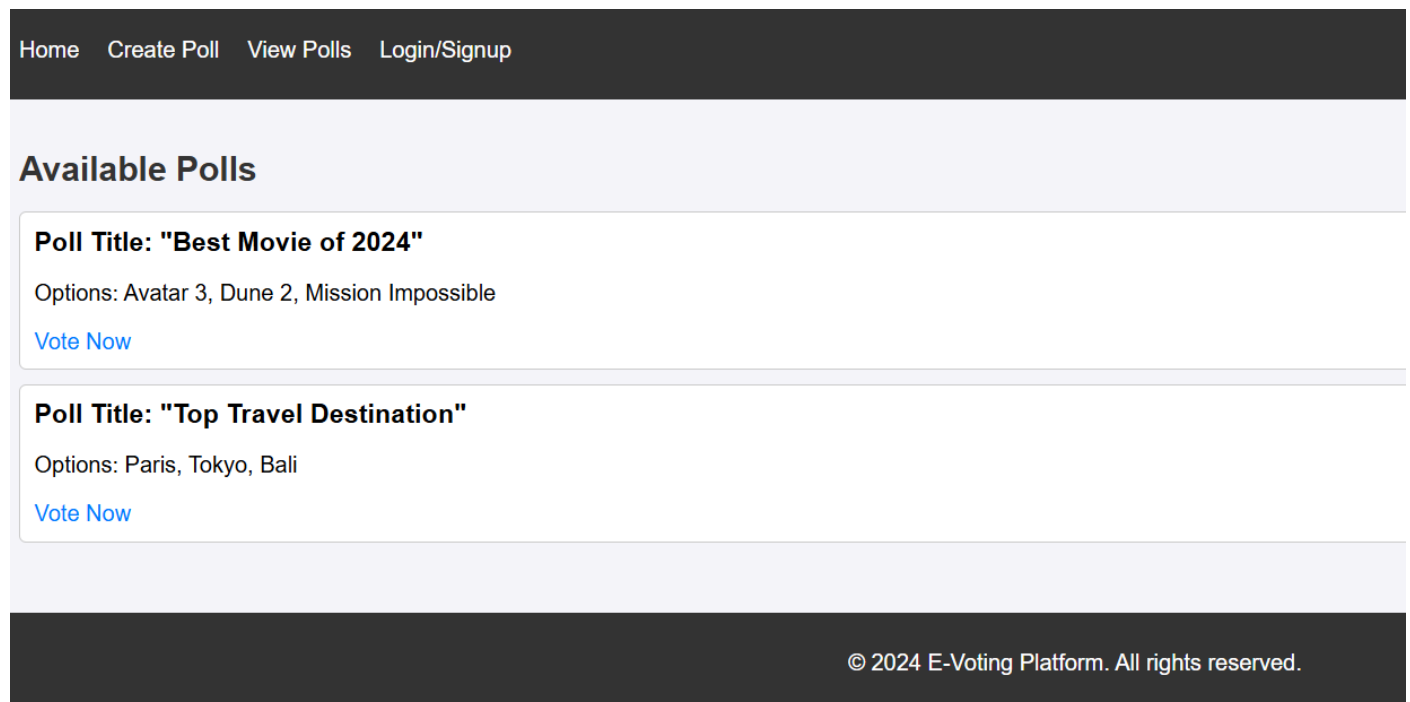


The screenshot shows the 'Create a Poll' page of an e-voting platform. At the top is a dark navigation bar with links: Home, Create Poll, View Polls, and Login/Signup. The main content area has a light blue header 'Create a Poll'. Below it, there is a 'Poll Title:' label followed by a text input field with the placeholder 'Enter your poll title'. Then, there is an 'Options (comma-separated):' label followed by a text input field with the placeholder 'Option1, Option2, Option3'. At the bottom of the form is a 'Create Poll' button. A dark footer bar at the bottom contains the copyright notice: '© 2024 E-Voting Platform. All rights reserved.'

Fig 5.2 Poll Creation page

5.4 Voting System

Users can cast votes on available polls. Each vote is recorded with the user's ID, poll ID, and choice.



The screenshot shows the 'Available Polls' page of an e-voting platform. At the top is a dark navigation bar with links: Home, Create Poll, View Polls, and Login/Signup. The main content area has a light blue header 'Available Polls'. Below it, there are two poll cards. The first card has the title 'Poll Title: "Best Movie of 2024"' and options 'Options: Avatar 3, Dune 2, Mission Impossible', with a blue 'Vote Now' link. The second card has the title 'Poll Title: "Top Travel Destination"' and options 'Options: Paris, Tokyo, Bali', also with a blue 'Vote Now' link. A dark footer bar at the bottom contains the copyright notice: '© 2024 E-Voting Platform. All rights reserved.'

Fig 5.4 Voting System page

5.5 Results Display

Poll results are calculated dynamically and displayed in real-time. But it will update the result soon

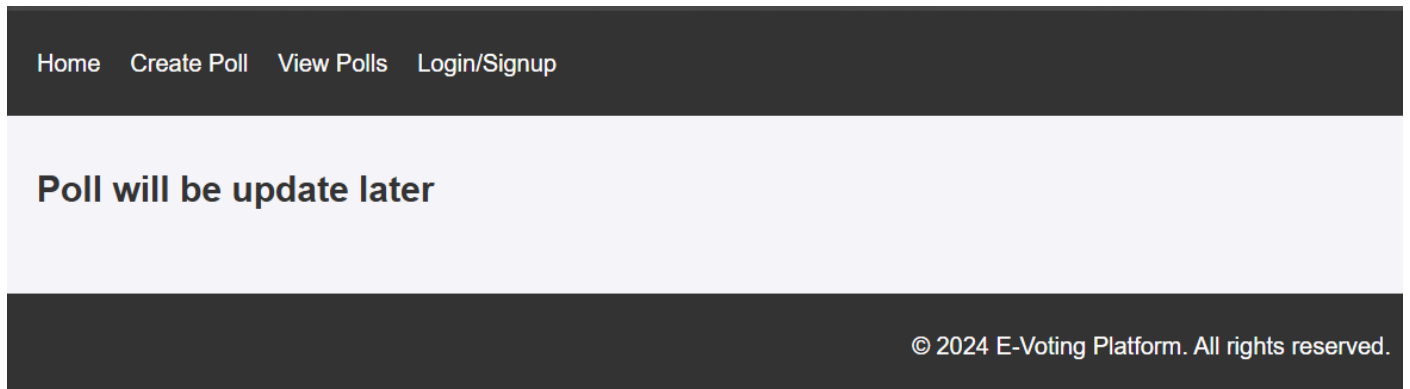


Fig 5.5 Result page

Chapter 6: Testing and Validation

6.1 Functional Testing

- Tested user registration and login functionality.
- Verified poll creation and vote submission processes.

6.2 Performance Testing

- Simulated multiple users to test system scalability.
- Measured response times for database queries.

6.3 Security Testing

- Validated encryption of passwords.
- Tested resistance to SQL injection attacks.

Chapter 7: Conclusion

7.1 Summary

This project successfully demonstrates the design and implementation of a secure and efficient E-Voting platform. By leveraging modern technologies, the platform ensures ease of use and robust data protection.

7.2 Future Scope

- Integrating mobile support for better accessibility.
- Enhancing security with two-factor authentication.
- Adding features for advanced poll analysis.

7.3 Limitations

- Requires internet access for usage.
- Relies on SQL Server for data storage, limiting portability to other database systems.